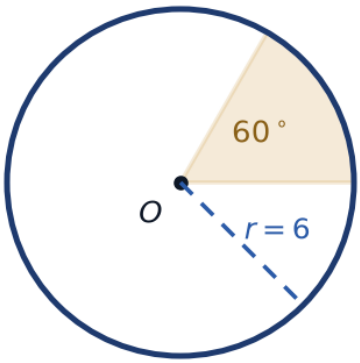
Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

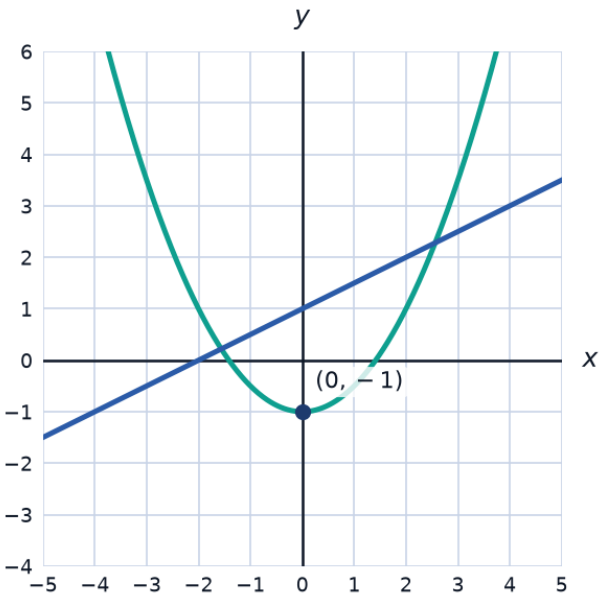
1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the modulus and principal argument of $z = -2\sqrt{3} + 2i$, and express z in polar form $r(\cos\theta + i\sin\theta)$.



2 **[Foundational]** • **Target: Powers of i**

Evaluate each power of i, using $i^2 = -1$ and the fact that powers of i repeat every 4:



a i^3 **b** i^4 **c** i^7 **d** i^{22}

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3 [Foundational] • Target: Powers of i

Evaluate i^{101}.

4 [Foundational] • Target: Addition and subtraction

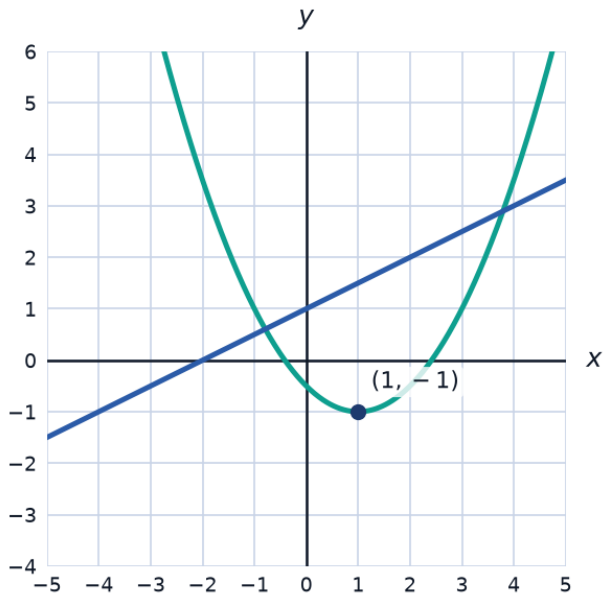
Let $z_1 = 3 + 4i$ and $z_2 = -2 + 5i$. Find $z_1 + z_2$.

5 [Foundational] • Target: Addition and subtraction

Let $z_1 = 2 - 3i$ and $z_2 = -1 - 4i$. Find $z_1 + z_2$.

6 [Foundational] • Target: Addition and subtraction

Find $(7 + 2i) - (3 - 5i)$.



7 [Foundational] • Target: Addition and subtraction

Find $(4 - i) - (-2 + 6i)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Multiplication

Multiply $(2 + 3i)(1 - 4i)$.

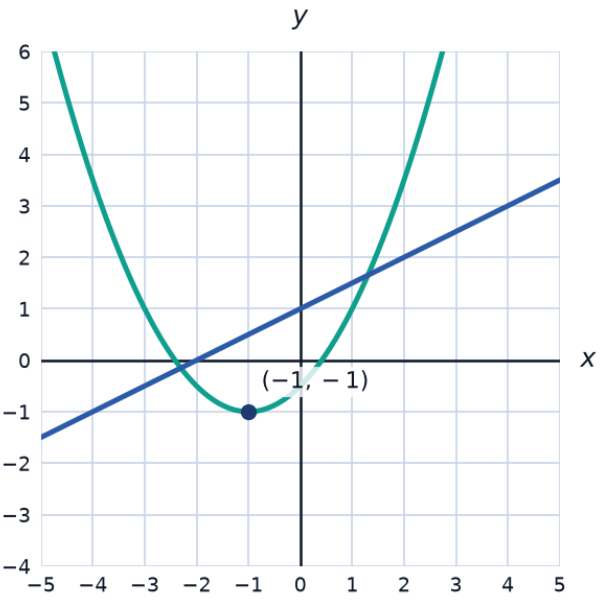
9 [Intermediate] • Target: Multiplication

Multiply $(-1 + 2i)(3 + i)$.

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10 [Intermediate] • Target: Multiplication

Multiply $(5 - 2i)(5 + 2i)$, and explain why the result has no imaginary part.



11 [Intermediate] • Target: Conjugate and modulus

For $z = 6 - 8i$, find $\bar{z}$ and $|z|$.

12 [Intermediate] • Target: Conjugate and modulus

For $z = -5 + 12i$, find $\bar{z}$ and $|z|$.

13 [Intermediate] • Target: Conjugate and modulus

Show that $z + \bar{z}$ is always real for $z = a + bi$.

14 [Intermediate] • Target: Division

Simplify $\frac{3 + 4i}{1 - 2i}$ by multiplying top and bottom by the conjugate of the denominator.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

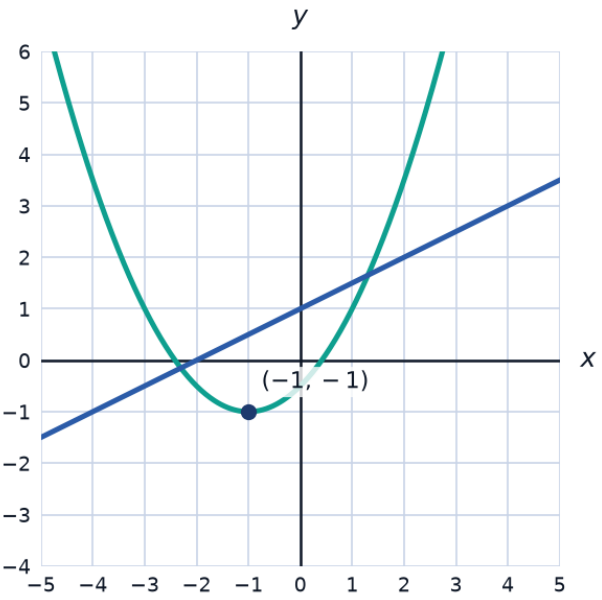
15 [Challenging] • Target: Division • Cross-Lesson: 6.01 Limits

Simplify $\frac{1 + i}{1 - i}$.

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16 [Challenging] • Target: Division • Cross-Lesson: 2.01 Matrices & Vectors

Simplify $\frac{7-i}{3+2i}$.



17 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 3.01 Sequences & Series

Solve $x^2 + 9 = 0$.

18 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 4.01 Advanced Functions & Equations

Solve $x^2 - 6x + 13 = 0$ using the quadratic formula.

19 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Solve $2x^2 + 2x + 5 = 0$ using the quadratic formula.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Limits

Simplify $\frac{3+4i}{1-2i}$ by multiplying top and bottom by the conjugate of the denominator.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Powers of i

Evaluate i^{130}, using the fact that powers of i repeat every 4.

a. i

b. $-i$

c. 1

d. -1

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22 [Foundational] • Target: Addition and subtraction

Evaluate i^{211}, using the fact that powers of i repeat every 4.

a. 1

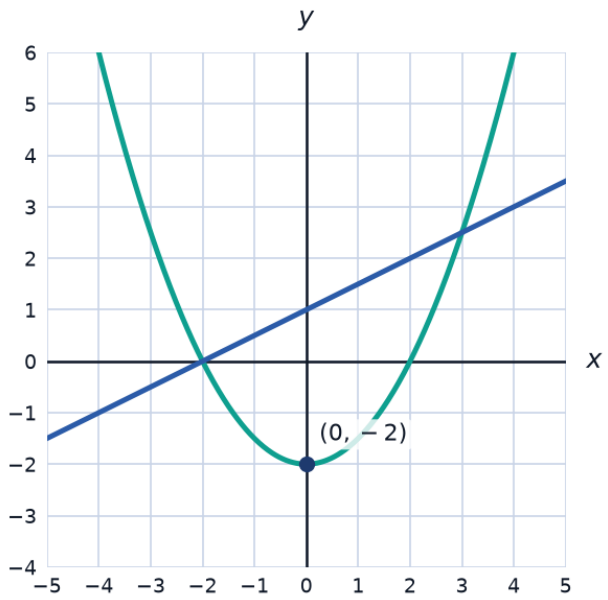
b. -1

c. $-i$

d. i

23 [Foundational] • Target: Multiplication

Evaluate i^{229}, using the fact that powers of i repeat every 4.



a. 1

b. i

c. -1

d. $-i$

24 [Foundational] • Target: Conjugate and modulus

Evaluate i^{206}, using the fact that powers of i repeat every 4.

a. i

b. $-i$

c. -1

d. 1

25 [Foundational] • Target: Division

Evaluate i^{62}, using the fact that powers of i repeat every 4.

a. i

b. -1

c. 1

d. $-i$

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26 [Foundational] • Target: Quadratic equations with complex roots

Evaluate i^{101}, using the fact that powers of i repeat every 4.

a. $-i$

b. -1

c. i

d. 1

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate i^{31}, using the fact that powers of i repeat every 4.

a. -1

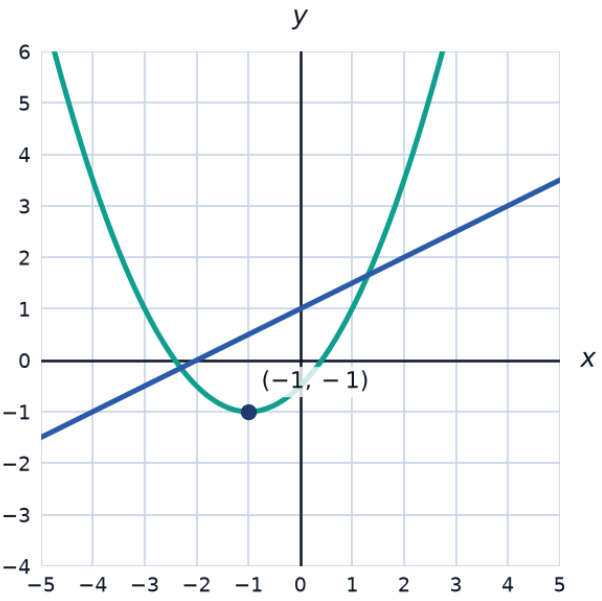
b. i

c. 1

d. $-i$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate i^{178}, using the fact that powers of i repeat every 4.



a. i

b. 1

c. -1

d. $-i$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Multiply $(3 - i)(-3 - 9i)$.

a. $-18 - 24i$

b. $-18 - 30i$

c. $-24i$

d. -18

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Multiply $(7 + 2i)(4)$.

a. $29 + 8i$

b. $28 + 8i$

c. $28 - 8i$

d. $28 + 28i$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Multiply $(7 - 6i)(4 + 8i)$.

a. $-20 + 32i$

b. $76 + 80i$

c. $76 - 20i$

d. $76 + 32i$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Multiply $(-3 - 7i)(1 - 3i)$.

a. $-24 + 18i$

b. $-24 + 2i$

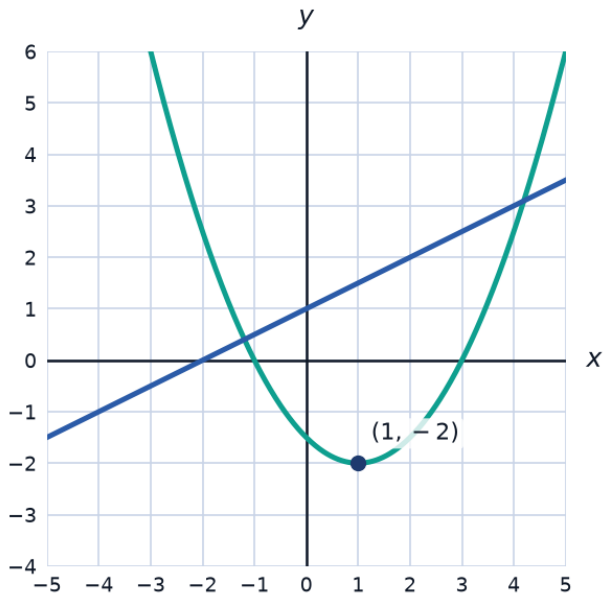
c. $18 + 2i$

d. $-24 + 16i$

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33 **[Intermediate]** • **Target: Powers of i**

Multiply $(8 - 5i)(4 + 6i)$.



a. $62 + 2i$

b. $62 + 68i$

c. $62 + 28i$

d. $2 + 28i$

34 **[Intermediate]** • **Target: Addition and subtraction**

Multiply $(2 - 2i)(-6 - 5i)$.

a. $-22 - 2i$

b. $-22 + 2i$

c. $-2 + 2i$

d. $-22 - 22i$

35 **[Intermediate]** • **Target: Multiplication**

Multiply $(-7 - 6i)(2 + 3i)$.

a. $4 - 9i$

b. $4 - 33i$

c. $4 - 32i$

d. $-32 - 33i$

36 **[Intermediate]** • **Target: Conjugate and modulus**

Simplify $\frac{6 + 8i}{2 + 8i}$ by multiplying top and bottom by the conjugate of the denominator.

a. $-8/17 + 19/17i$

b. $19/17 + 8/17i$

c. $19/17 - 8/17i$

d. $36/17 - 8/17i$

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37 [Intermediate] • Target: Division

Simplify $\frac{16+4i}{4+2i}$ by multiplying top and bottom by the conjugate of the denominator.

a. $18/5 - 4/5i$

b. $-4/5 + 18/5i$

c. $18/5 + 4/5i$

d. $23/5 - 4/5i$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 4.01 Advanced Functions & Equations

For $z = 9 - 40i$, find $|z|$.

a. 49

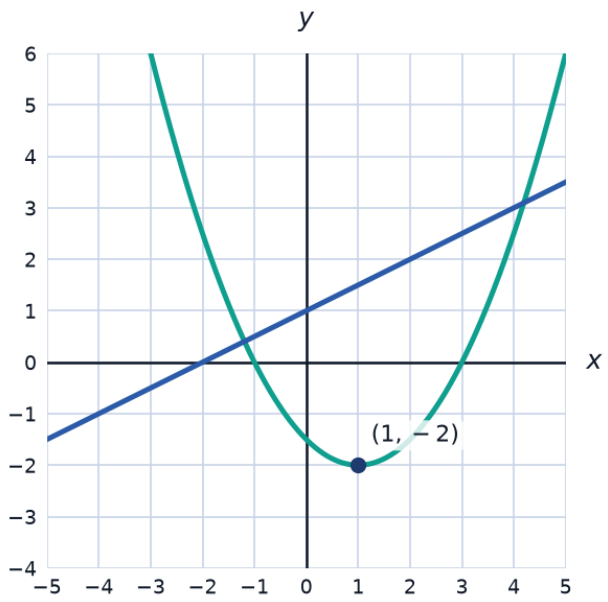
b. 42

c. 40

d. 41

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

For $z = 5 + 12i$, find $|z|$.



a. 17

b. 12

c. 14

d. 13

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Limits

For $z = 8 - 15i$, find $|z|$.

a. 17

b. 18

c. 16

d. 23

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Matrices & Vectors

Solve $x^2 - 14x + 113 = 0$ using the quadratic formula.

a. $7 + 8i$

b. $-7 + 8i,\ -7 - 8i$

c. $8 + 8i, 8 - 8i$

d. $7 + 8i, 7 - 8i$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Sequences & Series

Solve $x^2 + 14x + 85 = 0$ using the quadratic formula.

a. $-7 + 6i$

b. $-7 + 6i,\ -7 - 6i$

c. $-6 + 6i,\ -6 - 6i$

d. $7 + 6i, 7 - 6i$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Advanced Functions & Equations

Simplify $\frac{6+8i}{2+8i}$ by multiplying top and bottom by the conjugate of the denominator.

a. $-8/17 + 19/17i$

b. $19/17 + 8/17i$

c. $19/17 - 8/17i$

d. $36/17 - 8/17i$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

What is the polar (trigonometric) form of the complex number $z = -2 + 2i\sqrt{3}$?

a. $4(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

b. $4(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

c. $2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

d. $4(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

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45 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 6.01 Limits**

What is the real part of the complex number $(2 - 2i)^4$?

a. -64

b. 64

c. 0

d. -32

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